

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	3 October 2026
Team ID	SWTID-2026-4228
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The deliverable shall include the architectural diagram and the information as per Table-1 and Table-2.

Architecture Diagram: Employee Spreadsheet Import using ServiceNow Import Sets & Transform Maps

Source Sample Spreadsheet.xlsx Employee ID • Name • Email • Department • Location	ServiceNow Import Set Employee Import Staging / source records	Transform Map Sample Spreadsheet Import Field mapping + Coalesce
Target Employee Test u_employee_test Employee records	Validation Transform History Review imported / ignored records	Analytics Department • Location • Employee List Employee Analytics Dashboard

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	User interacts with ServiceNow through the web-based platform to configure tables, import sets, transform maps, reports and dashboards.	ServiceNow Web UI / Browser
2	Application Logic-1	Import Set logic stages spreadsheet data before transformation.	ServiceNow Import Sets
3	Application Logic-2	Transform Map logic maps source fields to target fields and creates or updates target records.	ServiceNow Transform Maps
4	Application Logic-3	Coalesce logic matches Employee ID to avoid duplicate records and update existing records.	Transform Map Coalesce
5	Database	Stores the target employee records and the custom Employee Test table.	ServiceNow Table / Glide Database
6	Cloud Database	Project data is stored within the ServiceNow platform instance; no separate cloud database is specified.	ServiceNow Platform Database
7	File Storage	Spreadsheet source is prepared externally and downloaded as Sample Spreadsheet.xlsx before import.	Excel / XLSX file

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Table-1 : Components & Technologies (continued):

S.No	Component	Description	Technology
8	External API-1	No external API is specified in the project solution.	Not specified
9	External API-2	No second external API is specified in the project solution.	Not specified
10	Machine Learning Model	The project does not specify a machine learning model.	Not specified
11	Infrastructure (Server / Cloud)	Application is deployed and operated in a ServiceNow instance; the project uses the ServiceNow platform for configuration, data import, transformation and reporting.	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No open-source application framework is specified in the project document. The solution is configured on the ServiceNow platform.	ServiceNow platform
2	Security	Security and access-control details are not explicitly specified. Access is governed through the ServiceNow platform and its configured application environment.	ServiceNow access controls
3	Scalable Architecture	The solution uses modular platform components: Import Set, Transform Map, target table, reports and dashboard. Additional employee data can be loaded through the same import process.	ServiceNow Import Sets & Transform Maps
4	Availability	The project operates within a ServiceNow platform instance. Specific load-balancer or distributed-server configuration is not specified.	ServiceNow platform
5	Performance	Transform Maps automate field mapping and transformation, while Coalesce prevents unnecessary duplicate records during repeated imports. Specific requests/sec or caching targets are not specified.	Transform Maps / Coalesce

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Technology Stack Summary

Layer	Project Technology / Component
Source Data	Sample Spreadsheet.xlsx — Employee ID, Name, Email, Department, Location
Staging	ServiceNow Import Set — Employee Import
Transformation	ServiceNow Transform Map — Sample Spreadsheet Import
Duplicate Handling	Coalesce on Employee ID
Target Data	ServiceNow custom table — Employee Test (u_employee_test)
Reporting	Employees by Department; Employees by Location; Employee List Report
Dashboard	Employee Analytics Dashboard

References:

1. ServiceNow project source: Import Data using Transform Maps (Spreadsheet).
2. Technology Stack template supplied for Project Design Phase-II.
3. ServiceNow platform concepts used in the project: Import Sets, Transform Maps, Coalesce, Reports and Dashboards.